

Evidence-Grounded Auditing of Identification Assumptions in Climate-Policy Causal Evaluations

Anonymous ACL submission

Abstract

Difference-in-differences (DID) studies are widely used to evaluate climate policy, yet their identification assumptions are more often *asserted* than *audited*. Because the counterfactual is unobserved, we ask instead whether each assumption is adequately supported by the evidence a paper reports. We introduce ARGUS, an end-to-end, *bounded-agentic* auditing system in which an LLM runs behind a fixed, deterministic pipeline and scores DID studies on an 11-dimension assumption–implication–evidence rubric, retrieving evidence, judging its adequacy, and producing a risk report rather than a causal verdict. Under realistic flaw injection (seeding one known error into an otherwise credible study) adequacy reasoning detects flaws that keyword matching cannot (0.18 versus up to 1.00 detection, all at zero false alarms), and a compute-graded ablation attributes the gain to per-dimension decomposition, with the retrieval gate trading detection for calibrated abstention. Across 27 real papers evidence grounding emerges as the coverage bottleneck, and a pilot diagnostic study reveals a systematic over-severity that a lightweight calibration layer partly corrects. ARGUS is thus a high-sensitivity screening tool that surfaces weakly evidenced identification assumptions for expert review.

1 Introduction

Difference-in-differences (DID) is widely used to evaluate climate and environmental policy, including carbon pricing, emissions-trading schemes, and emission standards. The credibility of these estimates depends on identification assumptions such as parallel trends, no anticipation, appropriate treatment of staggered adoption, and limited interference between units, each of which a decade of methodological work has shown to be fragile (Bertrand et al., 2004; Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfœuille,

2020; Roth et al., 2023). These assumptions are often difficult to assess in climate settings, where policies overlap, implementation is staggered, spillovers cross administrative boundaries, and emissions outcomes are imperfectly measured (Appendix A). Because environmental policy increasingly rests on this empirical evidence, causal conclusions built on weakly supported assumptions carry real costs for climate governance.

Yet applied papers often assert these assumptions rather than audit whether the reported evidence supports them, one facet of a broader reliability gap in empirical research (Ioannidis, 2005; Silberzahn et al., 2018; Brodeur et al., 2020). A paper may assert parallel trends over an event-study figure, leaving the reader to judge whether the design, pre-period estimates, inference, and policy timing justify the claim. Performing this assessment manually is slow, expert-intensive, and difficult to standardize.

We argue that identification credibility is auditable even when the true causal effect is not, building on the research-design tradition in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017; Rambachan and Roth, 2023). Because the counterfactual is unobserved, ARGUS does not ask whether an estimated effect is true. Instead, it asks whether each identifying assumption is adequately supported by the evidence reported in the paper, and where that support is weakest. ARGUS (named for Argus Panoptes, the hundred-eyed watchman of Greek myth) represents a DID study through an 11-dimension *assumption–implication–evidence* rubric and uses a bounded-agentic pipeline to retrieve relevant evidence, assess its adequacy, and produce an evidence-grounded risk report for expert review (Figure 1). Model agency is confined to scoped retrieval and assessment tasks behind a fixed, deterministic control flow, unlike free-roaming agents whose con-

trol flow is model-decided (Yao et al., 2023); ARGUS flags potential weaknesses rather than adjudicating causal truth.

Evaluating such an auditor is difficult because the true policy effect is unobserved. We therefore use *flaw injection* as local ground truth: starting from a supported study fixture, we introduce one known identification flaw and test whether ARGUS detects it, remains quiet on the clean version, and localizes the affected dimension. We complement this controlled evaluation with deployment on 27 real DID papers and a small expert-labelled diagnostic pilot used to identify calibration and applicability failures.

Contributions. Our contributions are:

1. a machine-auditable *assumption-implication-evidence* rubric for DID identification, spanning eleven dimensions (§3);
2. a bounded, evidence-grounded auditing pipeline with explicit abstention, separating retrieval uncertainty from substantive identification risk;
3. a controlled evaluation protocol built on planted identification errors (flaw injection), with same-model ablations and real-paper deployment (§5);
4. a diagnostic human study that reveals, and partially corrects, a systematic over-severity bias.

2 Related Work

DID identification and its threats. A large econometrics literature formalizes when DID estimators fail under staggered timing and heterogeneous effects: the “forbidden” comparisons of two-way fixed effects (Goodman-Bacon, 2021), negative weights (de Chaisemartin and D’Haultfœuille, 2020), robust estimators (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021), assumption and pre-trend surveys (Roth et al., 2023), and fragile inference under serial correlation (Bertrand et al., 2004). ARGUS adds no estimator: it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

LLM causal reasoning and benchmarks. A parallel line asks whether LLMs reason about causality. Benchmarks (CLadder (Jin et al., 2023), Corr2Cause (Jin et al., 2024), CausalBench (Zhou et al., 2024)) and critiques (Kiciman et al., 2024; Yang et al., 2024; Zečević et al., 2023) find models often reproduce causal *language* without reliable causal *inference*; a few move toward applied settings and toward separating *identification* from *estimation* (Lee et al., 2025; Sawarni et al., 2026). ARGUS asks the inverse question: are a *human* paper’s identification assumptions adequately evidenced? A causal verdict is out of scope by design.

Responsible AI for empirical research. AI is increasingly applied to research reliability: reproducibility checking, statistical-reporting validation, and AI-assisted review, motivated by wide analyst-driven result spreads (Silberzahn et al., 2018; Botvinik-Nezer et al., 2020) and specification searching in causal economics (Brodeur et al., 2020). LLMs are also used directly as evaluators and reviewers, judging model outputs (Zheng et al., 2023), giving feedback on research papers (Liang et al., 2024), and reviewing inside automated-science pipelines (Lu et al., 2024); these systems produce open-ended critique, whereas ARGUS scores a fixed rubric against in-paper evidence, so its outputs are comparable across papers and measurable against injected ground truth. ARGUS differs in target and posture: it audits *identification credibility* specifically (the part of an argument that licenses a causal claim) and issues no verdicts, producing evidence-grounded, human-reviewable reports that localize risk for the expert’s final call.

NLP for climate evidence. Climate-NLP systems verify claims against evidence (CLIMATE-FEVER (Diggelmann et al., 2020)), detect environmental claims (Stammach et al., 2023), analyse corporate climate disclosures with traceable LLM answers (Ni et al., 2023), and tune models for faithful evidence-based question answering (Schimanski et al., 2024). These systems ground answers in documents; ARGUS extends document-grounded verification from *claims* to the *identification argument* of a causal study, auditing whether reported evidence supports each assumption.

Evaluation methodology and bounded agents. Our evaluation follows benchmarking lessons:

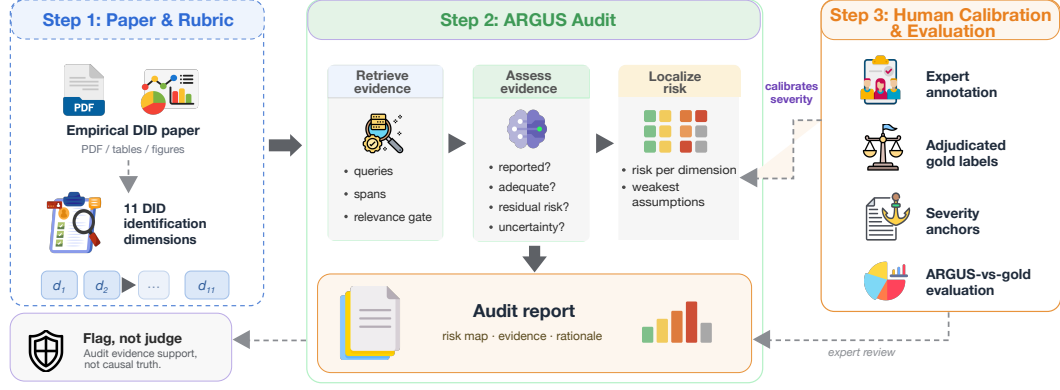


Figure 1: Overview of ARGUS. Step 1 converts an empirical DID paper into an auditable structure using an 11-dimension identification rubric. Step 2 retrieves dimension-specific evidence, assesses reported-evidence adequacy with a bounded LLM audit module, and localizes risk in a human-reviewable report. Step 3 uses expert annotation and adjudicated gold labels to evaluate and calibrate audit severity offline; deployment requires only expert review of the report. ARGUS flags dimensions for expert review.

holistic evaluation (Liang et al., 2023) over a single number (we report a layered retrieval/relevance/adequacy/risk profile); dynamic, adversarial benchmarking (Kiela et al., 2021) (our model-in-the-loop flaw injection); and the caution that operationalizing a capability is non-neutral (Raji et al., 2021; Gebru et al., 2021) (§6). Where ReAct-style agents (Yao et al., 2023) act over an open horizon, ARGUS *bounds* the paradigm (agency confined to two scoped sub-tasks behind deterministic control flow), trading autonomy for reproducibility and evidence-traceability.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA/spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it. For a carbon-pricing or emissions-trading evaluation, for instance, the parallel-trends dimension expects an event-study or pre-trend test with null pre-period coefficients, while no-anticipation turns on the gap between a scheme’s *announcement* and *implementation* dates. The rubric

is declarative, stored in a configuration file (`identification_dimensions.yaml`), so the audited dimensions are explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded-agentic pipeline

The audit runs a fixed, deterministic sequence, *decomposition* → *extraction* → *assessment* → *localization* → *report*, shown in the top band of Figure 2. The stage order is written in code and never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand; *extraction* gathers, per dimension, the in-paper evidence relevant to that assumption; *assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence; *localization* aggregates these judgements into a risk map ranking where support is weakest; and *report* renders a transparent, evidence-grounded document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps: extraction and assessment. Each invokes a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call: `evidence_search` over the paper’s text, figures and tables, `figure_parse` for structured figure/table content, and `policy_lookup` for

policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (`max_steps`) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop. In the configuration evaluated in §5 the step budget is minimal (`max_steps=1`), and the LLM assessment path resolves to a fixed per-dimension sequence (deterministic retrieval, one relevance-gate call, one adequacy call). The evaluated system is therefore closer to structured, dimension-scoped prompting over retrieved evidence than to an open-ended agent; for an audit, we treat this near-determinism as a feature.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, undermining the central evaluation claim of §4: that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable metrics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk; the judgement itself stays with the human.

3.3 Output

ARGUS emits study–dimension–risk triplets, each with an evidence-grounded rationale, alongside the human-readable report. It never produces a verdict on the true causal effect. This output shape keeps the system both auditable by a human and measurable by the evaluation that follows.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest (the true causal effect of a carbon price, an emissions cap, or an emission standard) is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a credible policy-evaluation fixture, modelled on evaluations of China’s provincial emissions-trading pilots (announced 2011, rolled out 2013–2014), we inject

exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`). Because we control which dimension we damaged, we know the correct audit outcome for that dimension, without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension above a threshold?
- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

4.2 Realistic perturbation, not sentinels

How a flaw is injected is a methodological choice that matters. A naïve injector appends a sentence that *names* the flaw: e.g. “treated and control have non-zero pre-period coefficients with significant leads.” If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector matches on. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws *remove* the section, figure, or table that supplies the support for a dimension (e.g. deleting the placebo-robustness section or the balance table), modelling a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered policy timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regression test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection is thereby a genuine task rather than a string match.

5 Results

We evaluate ARGUS along four axes: a synthetic flaw-injection protocol, an architecture ablation,

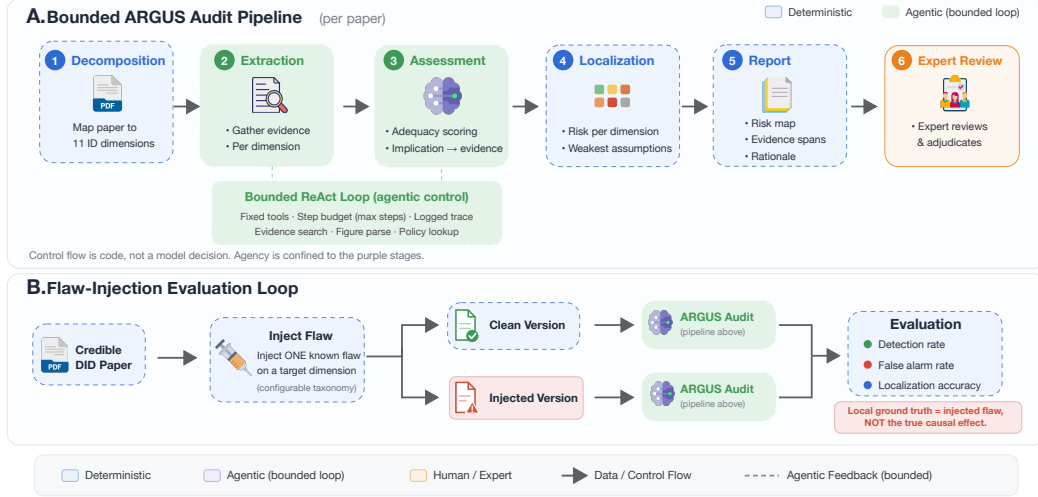


Figure 2: ARGUS architecture. The top band is the audit pipeline, a fixed deterministic sequence; agency is confined to the two agentic stages (extraction and assessment), each a bounded ReAct loop with a fixed tool set (evidence search, figure parse, policy lookup), a hard step budget, and a logged trace. The bottom band shows the flaw-injection evaluation loop, which injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth, never against the unobserved true effect.

a run over 27 published difference-in-differences (DID) papers that stress-test the audit architecture across empirical economics, beyond the climate-policy setting (carbon pricing, emissions trading, emission standards) that defines its target deployment, and a calibrated human-gold pilot. Four findings emerge: (i) judging evidence *adequacy*, not keyword presence, is what detects identification flaws; (ii) per-dimension decomposition drives detection, while the bounded retrieval gate turns retrieval failure into an auditable *unknown* signal; (iii) on real papers the dominant practical bottleneck shifts from causal reasoning to evidence grounding; and (iv) against a human gold ARGUS is systematically over-severe, a bias a lightweight calibration layer partly corrects.

5.1 Adequacy beats presence; architecture beats a single pass

We compare three assessors under one flaw-injection protocol on the `clean_supported` fixture (one flaw injected per run, all eleven flaws; `max_steps=1`). A deterministic **keyword baseline** scores each dimension from the presence of supporting and risk-indicating vocabulary. The **two-stage ARGUS assessor** judges evidence *adequacy* (whether the reported evidence actually establishes the assumption) over the *same retrieved evidence* as the keyword baseline, so that pair isolates the judgement policy. A **single-pass** abla-

Table 1: Synthetic flaw injection (`clean_supported`, eleven flaws; all LLM rows `gpt-4o`, replicated on the dated 2024-11-20 snapshot; see Appendix C on serving drift). Keyword and two-stage share the same retrieved evidence; the ablation arms share the model and rubric but remove retrieval, the gate, or both. The final row is the leaked, pre-fix circular setup.

Assessor	calls	detection	false alarm	localization
keyword baseline	0	0.182	0.000	0.182
single-pass	1	0.818	0.000	0.818
per-dimension, no retrieval	11	1.000	0.000	1.000
two-stage ARGUS	22	0.727	0.000	0.727
(circular, pre-fix)	—	1.000	0.000	1.000

tion removes the architecture entirely: the full paper and all eleven dimensions in one `gpt-4o` call, with the same model, temperature (the sampling-randomness setting, held at 0), risk schema, and scoring harness, but no staged retrieval, relevance gating, bounded tool use, or explicit *unknown* state. The only difference is architectural. Table 1 reports the headline metrics.

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm and correctly flags the two omission flaws (`measurement_break`, `spillover_contamination`) by the *absence* of their vocabulary, but beyond those detection is only 0.182 (Figure 3B). Two failure

modes explain it: *cross-section vocabulary bleed* (deleting the placebo section does not help when another section also mentions “placebo”) and blindness to *every* commission flaw, whose flawed-but-plausible evidence carries the expected topic vocabulary. Keyword presence is not evidence adequacy.

Reasoning over adequacy closes the gap. Swapping *only* the assessment policy (extraction unchanged, so both see the same evidence) to an LLM that judges adequacy raises detection from 0.182 to 0.727 with no false alarms, catching *every* commission flaw the keyword scorer is blind to; handing each dimension the full fixture (the per-dimension arm) reaches 1.000. The model reads flawed-but-natural prose the keyword scorer cannot; the audit report in Appendix E shows it flagging the injected anticipation flaw with the announcement-gap rationale, with no sentinel keyword present. This is the empirical core of the design: placing judgement where reasoning is required.

What the architecture buys. The gated pipeline detects *less* than its own ablations here (0.727 vs 0.818 and 1.000) by design: its three misses are all *omission* flaws on which retrieval finds nothing and it abstains rather than guessing. The gate’s value is behavioural and shows on real papers, where an ungated model almost never abstains (unknown 0.4% versus 38.9% over 275 common cells; Appendix B) and instead commits a substantive risk it cannot ground.

Scope. These are synthetic fixtures with few dimensions; ceilings on clearly injected flaws are a proof-of-concept, not a solved task. The sharper tests follow: subtler flaws (§5.2), then real papers (§5.3).

5.2 Robustness: a larger, harder variant benchmark

The eleven flaws above are deliberately clear-cut. To test whether the result survives subtler and more varied flaws, we built a second benchmark of 33 flaw variants: three per dimension (two *commission* rewrites and one *omission*), each verified to contain none of the keyword detector’s sentinel phrases. We audit this set with the two-stage AR-GUS assessor (gpt-4o) over three runs at temperature 0; verdicts are near-deterministic (32/33 identical across runs; Appendix C). Detection is 0.73–

0.76 and localization 0.64–0.70, in line with its behaviour on the eleven clear flaws.

Where the gain comes from. A compute-graded ablation on this benchmark (Appendix B) separates the pipeline’s ingredients: a full-paper single pass (one call) detects 0.727; per-dimension prompting without retrieval (eleven calls) detects 0.879; the full two-stage pipeline (twenty-two calls) detects 0.747. Task *decomposition* drives synthetic detection; the retrieval-and-gate layer then *trades* omission detection for abstention (on these small fixtures a gateless assessor sees deleted sections directly in its context; on real 30-page papers it instead commits judgements nearly everywhere, §5.3).

Cross-model robustness. Porting the unchanged pipeline across providers (Appendix B) shows the core result is not model-specific: *commission* detection is high for every model tested (gpt-4o 0.89, Claude Opus 4.8 1.00, Gemini 2.5 Flash 0.91, Llama 3.1 8B 0.95). What differs is the operating point: gpt-4o is the only conservative auditor (false alarm 0.09, localization 0.66), while the others buy sensitivity with alarms (false alarm 0.18–0.82, localization 0.24–0.36): the pipeline ports everywhere, its precision profile is model-dependent.

The split by flaw type is the informative result (Table 2). The assessor catches 0.86–0.91 of *commission* flaws (flawed-but-present evidence, the case adequacy reasoning targets) but only 0.45 of *omission* flaws. The omission misses are not confident errors: the six missed variants all abstain to *unknown* rather than assert low risk, because deleting the supporting section makes retrieval fail. This reproduces, under controlled injection, the same *evidence-grounding* bottleneck we find on real papers (§5.3): the pipeline reasons well over evidence it can retrieve and abstains when it cannot. The only false-alarm signal (0.09, identical across runs) is concentrated in *inference*, where the clean fixture’s section is itself flagged, the same over-severity we quantify against the human gold (§5.5). The “no false alarms” result in Table 1 is on the original, clearer eleven-flaw set; this harder benchmark surfaces the inference over-flag.

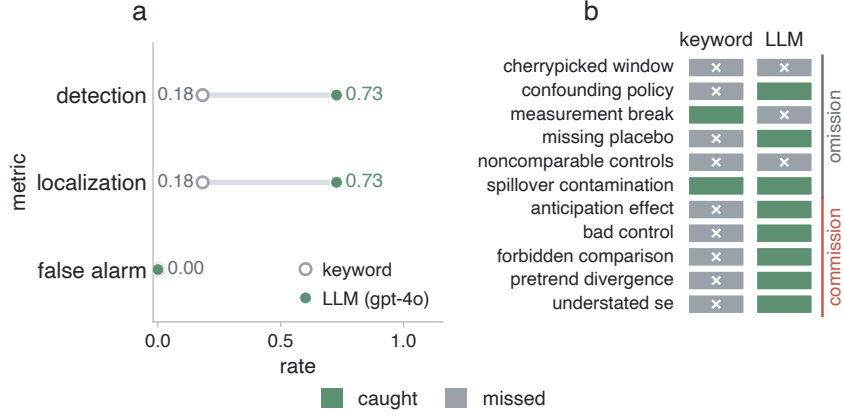


Figure 3: Keyword baseline vs. LLM evidence-adequacy assessor ($n=11$ flaws, realistic injection). (a) Swapping *only* the assessment policy (same retrieved evidence) recovers detection and localization $0.182 \rightarrow 0.727$ with false alarm 0. (b) Per-flaw caught/missed: the keyword baseline is *blind to every commission-type flaw*, all of which the LLM catches; the LLM’s three misses are omissions on which it abstains. Table 2 reports the harder 33-variant benchmark.

Table 2: Harder 33-variant benchmark (gpt-4o, mean of three runs at temperature 0; near-deterministic, 32/33 verdicts identical). Adequacy reasoning catches *commission* flaws; *omission* misses are abstentions to *unknown*, marking the evidence-grounding bottleneck rather than wrong judgements.

variant type	n	detection	localization
commission	22	0.89	0.82
omission	11	0.45	0.33
all	33	0.75	0.66

5.3 From synthetic fixtures to real papers: a relocated bottleneck

We next audit a corpus of 27 published DID papers drawn from top economics journals (Appendix D details its composition). There is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 5). The keyword baseline is permissive: 91% of its judgements are *low*. It sees topic vocabulary, declares the dimension supported, and flags almost nothing. The two-stage LLM is differentiated, and its most informative output is that $\sim 39\%$ of judgements are *unknown*: the relevance gate could not surface evidence, so the system says so rather than over-flagging. Two regimes appear: dimensions whose discussion is diffuse or lives in figures the text retrieval cannot reach (parallel trends, no anticipation, placebo) are dominated by *unknown*, while dimensions where a section is retrieved (sample period, concurrent policies, SUTVA) are judged inadequate. The finding is methodological:

moving to real papers is bottlenecked not by causal reasoning but by *evidence grounding*. The design therefore reports retrieval quality and an explicit *unknown* state distinct from a substantive high-risk judgement.

Judgements are grounded rather than pattern-matched (Appendix E gives a qualitative example). Roughly 45% of judgements on these top-journal papers are *high*; without dimension-level expert labels we cannot yet say whether this is warranted or over-strict where retrieval succeeds. We claim the contrast in behaviour and the localization of the grounding bottleneck, not the correctness of any individual flag.

5.4 A calibrated human gold (pilot)

To test whether ARGUS’s flags agree with expert judgement, we built a small expert gold: two annotators independently labelled risk on 5 DID papers $\times$ 11 dimensions (55 cells), with an adjudication round resolving disagreements (construction details and agreement statistics in Appendix F). Disagreement is almost entirely ordinal and concentrates on *non-canonical DID designs* (continuous-shock, event-study) where a dimension like staggered timing is not applicable as conventionally defined: a real share of “expert disagreement” is *design applicability*, not severity. The adjudicated gold (55 cells) is low 29, medium 24, high 2. This is a pilot ($n=5$, two annotators), read as preliminary.

5.5 ARGUS versus the human gold

We use this gold as a *diagnostic pilot*, not a validation of accuracy: with only two high-risk gold cells across five papers it cannot establish agreement with experts, but it is enough to localize *how* ARGUS fails (Table 7, Figure 4). ARGUS abstains on 22/55 cells, and when it answers its errors are almost entirely one-directional: 28/33 answered cells are more severe than the adjudicated gold, none less severe; gold contains two high-risk cells, ARGUS assigns 24. With so few high gold labels, precision and recall are noisy; the robust conclusion is over-sensitivity to missing or partial evidence: a *high-sensitivity screening tool* for prioritizing human review.

5.6 Calibration: from diagnosis to improvement

The gold comparison also points to a fix. An error analysis of the over-severe cells is sharp: of ARGUS’s over-severe *high* judgements, 23/24 have `evidence_status=missing` and 20/24 have `retrieval_quality=weak`. ARGUS treats “weak retrieval found nothing” as a substantive *high*, conflating a *retrieval failure* with the paper lacking the evidence, which the human, reading the full PDF, resolves to low/medium: a fixable calibration target.

An *oracle-evidence* check sharpens the diagnosis into two distinct constraints. An expert located supporting evidence for *all* 55 cells, so all 22 pipeline abstentions are retrieval failures, not reporting gaps: grounding limits *coverage*. But feeding the assessor the expert’s evidence spans does not restore agreement (50/55 judgements more severe than gold; exact 0.07): severity calibration, not evidence access, limits *agreement*.

A deterministic calibration layer over fields the LLM already produces cuts over-severity while holding high-risk detection unchanged on the pilot’s two high cells; the dominant rule demotes a weak-retrieval *high* to *medium* (keep coverage) or unknown (abstain). Keeping coverage raises exact agreement from 0.15 to 0.45; abstaining reaches weighted $\kappa = 0.25$ but leaves 13 answered cells (Appendix, Table 8, Figure 6). Calibration corrects severity but cannot recover missing evidence (correct *coverage* needs better grounding), so the human gold does more than evaluate ARGUS; it calibrates it.

6 Discussion

What ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it (Jacobs and Wallach, 2021; Raji et al., 2021). ARGUS audits whether a climate-policy study’s identification assumptions are *adequately supported by its reported evidence*: reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself. The two can diverge, so ARGUS surfaces risk for an expert to weigh and makes no ruling on the effect itself.

Why climate policy. The identification challenges above stress several assumptions at once, and climate-policy estimates feed directly into governance and resource allocation, so a confidently wrong credibility judgement is costly; ARGUS therefore abstains rather than guesses when it cannot ground the evidence.

Evidence grounding is the dominant practical constraint. A further gap separates the construct from the task: retrieving the relevant evidence. Our real-paper runs show it limits coverage. A reasoning assessor handed off-target evidence over-flags every dimension, so ARGUS reports a *retrieval-quality* status and an explicit unknown risk (“the system could not find the evidence”) distinct from a substantive high-risk judgement (“the paper lacks it”). This is why the pipeline separates retrieval, relevance filtering, and adequacy assessment.

7 Conclusion

The right target for AI assistance in empirical climate-policy research is not the unobservable causal effect but the *identification credibility* of the study that estimates it. ARGUS operationalizes this stance: a bounded pipeline decomposes identification into auditable dimensions and localizes risk in an evidence-grounded, human-reviewable report. Synthetic flaw injection supplies reproducible local ground truth without access to the true effect: a keyword baseline stays blind to flawed-but-plausible evidence that an adequacy-judging model recovers. A human-gold diagnostic pilot further shows that expert labels can do more than evaluate the auditor: they expose, and help correct, a systematic over-severity bias. Future work: expert-authored flaws, expert annotation at scale (the central open challenge), stud-

ies of whether reports help reviewers, and settings beyond the China fixture. Auditing identification credibility is, we argue, a tractable and responsible role for AI in empirical environmental-policy research.

Limitations

Our human gold is pilot-scale: 5 papers and 55 cells with two annotators and a single adjudication round, so the expert comparison is a preliminary signal rather than a calibrated benchmark. The current scope is confined to difference-in-differences identification; whether the rubric-plus-injection methodology transfers to other quasi-experimental designs is untested. The rubric, flaw taxonomy, and injector are co-designed, so synthetic detection rates are best read as internal consistency under known threats rather than external generalization; flaws authored by independent experts are future work. Relatedly, flaw injection certifies the auditor against the *injected* threat on a single dimension only and inherits the taxonomy’s blind spots, so we treat detection and localization as necessary rather than sufficient, pairing them with the false-alarm rate and the human-gold comparison. Evidence grounding is the dominant practical constraint on coverage (the oracle-evidence check in §5.6 shows severity calibration separately limits agreement, though its expert spans average only ~ 120 characters, so part of the residual over-severity reflects judging brief excerpts): when retrieval returns off-target passages the assessor cannot judge adequacy, which is why ARGUS emits an explicit unknown state rather than a substantive verdict. Finally, ARGUS is an auditor, not an oracle: it screens whether reported evidence supports a study’s identification assumptions and does not adjudicate the true causal effect, which is never observed. Its outputs are meant to surface risk for expert review, not to replace it.

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A Domain Motivation and Design Rationale

A.1 Why climate-policy DID is a demanding audit setting

Climate-policy evaluations present several recurring identification challenges. First, policy announcement, legal adoption, and actual implementation may occur at different dates, creating scope for anticipation. Second, carbon markets and environmental regulations are commonly introduced at different times across jurisdictions, making heterogeneous and staggered treatment effects central rather than exceptional. Third, emissions, investment, and production responses can spill across regional borders, challenging no-interference assumptions. Fourth, climate policies frequently overlap with industrial, energy, and air-pollution regulations, complicating the exclusion of concurrent interventions. Fifth, policy effects may emerge with long and heterogeneous lags. Finally, emissions and environmental outcomes are assembled from administrative, modeled, or remotely sensed sources whose measurement properties vary across places and time. These features do not make DID unique to climate policy, but they make the quality of identification evidence especially consequential for policy interpretation and comparison.

A.2 Identification auditing and research reliability

ARGUS addresses one component of a broader research-reliability problem: published conclusions can vary with specification choices, undisclosed analytic flexibility, and researcher degrees of freedom (Leamer, 1983; Ioannidis, 2005; Simmons et al., 2011; Gelman and Loken, 2014; Silberzahn et al., 2018; Brodeur et al., 2020; Botvinik-Nezer et al., 2020). This is the identification-credibility component of the broader research-design or “credibility revolution” program in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017), which increasingly treats an assumption like parallel trends as a matter of degree rather than a binary (Ramachan and Roth, 2023; Roth, 2022). ARGUS does not attempt to audit every component of an empirical paper; it focuses on the identification argument that licenses a causal interpretation.

Table 3: Compute-graded ablation on the 33-variant benchmark (gpt-4o). det / fa / loc = detection, false alarm, localization.

arm	calls/paper	det	fa	loc
full-paper single pass	1	0.727	0.000	0.727
per-dimension, no retrieval	11	0.879	0.000	0.879
two-stage (retrieval + gate)	22	0.747	0.091	0.657

A.3 Why ARGUS uses bounded agency

Open-ended agents may choose different tools and reasoning paths across runs, making their outputs difficult to reproduce and compare. Large language models are increasingly used for evaluative and reviewing tasks (Zheng et al., 2023; Liang et al., 2024; Lu et al., 2024), but as open-ended agents their reasoning traces are hard to audit. ARGUS instead fixes the pipeline order in code and confines model-driven decisions to evidence retrieval and adequacy assessment. Each model call operates under a fixed tool set, a hard step budget, and a logged trace. This design trades autonomy for reproducibility, inspectability, and controlled evaluation under flaw injection.

B Ablation Matrix and Cross-Model Panel

Table 3 grades compute on the 33-variant benchmark: task decomposition (one adequacy call per dimension) drives synthetic detection, and the retrieval-and-gate layer converts omission detection into abstention. Table 4 ports the unchanged pipeline across providers: commission detection is uniformly high, while false-alarm and localization profiles vary sharply by model. Cross-provider decoding defaults differ (newer reasoning models reject an explicit temperature), so these are single runs.

On real papers the gate’s behavioural contrast sharpens: over the 25 papers common to both runs (275 cells), the ungated single pass almost never abstains, converting the gated pipeline’s unknown cells into confident low (1.8% → 14.9%) or medium (14.2% → 43.6%) judgements while the high-risk share stays comparable (41.1% vs 45.1%).

C Reproducibility Details

All gpt-4o stages use temperature 0 with max_steps=1; newer reasoning models (Claude Opus 4.8) reject an explicit temperature and run

Table 4: Cross-model panel on the 33-variant benchmark (single run per model; gpt-4o is the three-run mean). comm/omit = commission/omission detection.

model	det	fa	loc	comm	omit
gpt-4o	0.75	0.09	0.66	0.89	0.45
Claude Opus 4.8	0.91	0.82	0.24	1.00	0.73
Gemini 2.5 Flash	0.79	0.18	0.36	0.91	0.55
Llama 3.1 8B (local)	0.94	0.27	0.27	0.95	0.91

at provider defaults. The identification rubric, flaw taxonomy, and 33-variant catalog are declarative configuration files in the code release, and the keyword baseline plus all leak-freedom (non-circularity) checks are deterministic and run without model access. Eleven-flaw numbers are July-2026 runs of the released runner; detection replicates exactly across the gpt-4o alias and the dated gpt-4o-2024-11-20 snapshot (localization varies between runs, 0.455–0.727). A June-2026 run of the same code had scored two-stage 1.000/0.909 and single-pass 0.455; neither value reproduces on current serving of the alias *or* the dated snapshot, indicating provider-side serving drift, so we report only currently replicable numbers. The 33-variant benchmark was run three times end-to-end (roughly 750 model calls per run, ~20 minutes and ~US\$8 per gpt-4o run; the local Llama run took ~67 minutes on a laptop at zero API cost); verdicts were near-deterministic (32/33 identical across runs), and Table 2 reports the three-run mean. The two-call-per-dimension pipeline audits one paper for roughly US\$0.25 of gpt-4o usage. The oracle-evidence check (§5.6) is 55 adequacy calls over expert-annotated spans.

D The Real-Paper Corpus

The 27 published DID studies audited in §5.3 are drawn from top general-interest and field economics journals (Table 5). By topic, the corpus spans finance and banking (5), environment and energy (5), health (4), education (4), international trade (3), labour (3), crime and social policy (2), and industrial location (1); the environment-and-energy subset covers industrial-activity effects of environmental regulation, air quality and early-life mortality, and energy pricing. Audit behaviour is not driven by domain: the environment-and-energy subset’s risk distribution (high 0.40, unknown 0.38; 55 cells) is close to the remainder’s (high 0.46, unknown 0.39; 242 cells). Paper-level identities are withheld here

Table 5: Journal composition of the 27-paper real corpus.

journal	papers
Quarterly Journal of Economics	7
American Economic Review	6
Journal of Political Economy	4
Review of Economics and Statistics	4
Journal of Finance	3
AEJ: Economic Policy / J. Monetary Econ. / ILRR	3

Table 6: ARGUS audit report (abridged) on the fixture with the injected anticipation flaw. Every row cites the evidence it rests on; the injected dimension is the one flagged high.

dimension	risk	rationale (abridged)
no anticipation	high	outcomes began changing between the policy announcement and its rollout, undermining no-anticipation
inference	medium	clustering and wild-cluster bootstrap reported, but the number of clusters is not stated
parallel trends	low	event-study with null pre-period coefficients and placebo tests
treatment timing	low	heterogeneity-robust estimators with a Goodman–Bacon decomposition
7 further dimensions	low	supporting evidence retrieved and judged adequate

to preserve double-blind review of the associated dataset and will be listed in the camera-ready version.

E Qualitative Audit Examples

A complete audit report. Table 6 shows ARGUS’s actual output on the emissions-trading fixture with the *anticipation* flaw injected (§4): the report flags exactly the damaged dimension as high, giving the anticipation-gap rationale, keeps nine supported dimensions at low, and shows a mild over-severity trace on *inference* (the fixture does not state the number of clusters), the same bias the human-gold pilot quantifies. This is the dimension-level, evidence-cited report a reviewer receives.

A grounded judgement on a real paper. On a trade-and-wages paper ARGUS retrieves the employment-results section and flags *inference* as high because “the paper only mentions heteroskedastic-consistent standard errors and does not provide information on clustering or the number of clusters”: a specific econometric observation, not a keyword hit. Elsewhere it distinguishes a *related but different* reported check (an alternative-specification section) from the placebo test an assumption actually requires.

Table 7: ARGUS-LLM vs adjudicated human gold (55 cells, 5 papers) under two unknown policies: coverage (drop abstentions) and strict (unknown counts as mismatch). Pilot-scale: only 2 gold cells are *high*; the robust signal is the one-directional over-severity and the abstention rate.

unknown policy	<i>n</i>	exact	κ	wt. κ
exclude (coverage)	33	0.152	0.019	0.044
mismatch (strict)	55	0.091	0.002	—

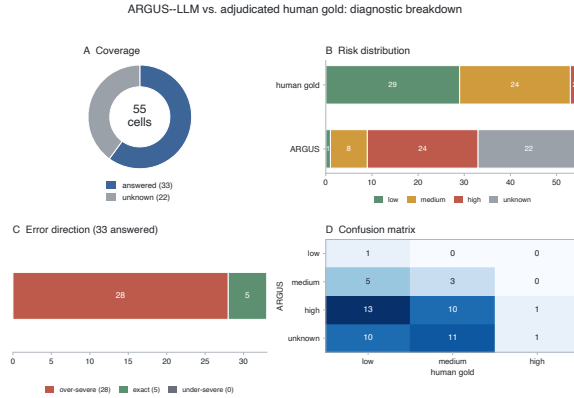


Figure 4: ARGUS-LLM versus adjudicated human gold on 55 dimension-level cells (5 papers). (a) coverage, separating answered judgments from abstentions; (b) severity distributions of human gold and ARGUS; (c) error direction among the 33 answered cells; (d) the ARGUS-by-gold confusion matrix. The dominant signal is one-directional over-severity: ARGUS assigns higher risk than gold in 28/33 answered cases and lower risk in none.

F Supplementary Figures and Tables

Pilot-gold construction (§5.4): round-1 agreement on the raw 3-level scale was low (Cohen’s $\kappa = 0.17$) but almost entirely ordinal (weighted $\kappa = 0.31$; 96% of disagreements are a single severity step). Of the 27 round-1 disagreements, 16 resolved to a single label once the low/medium boundary was anchored, and 11 were flagged intrinsically ambiguous.

This appendix collects supporting material referenced from the main text: the per-dimension risk distribution on the 27-paper real corpus (Figure 5), and the deterministic calibration results against the adjudicated human gold (Table 8, Figure 6), and the agreement metrics for ARGUS versus the human gold (Table 7).

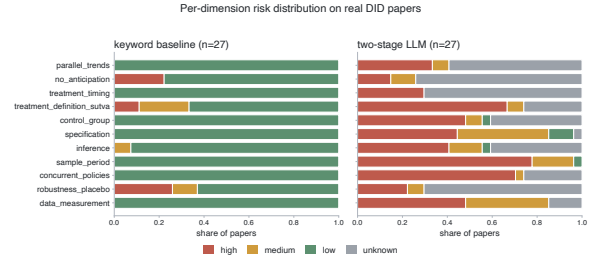


Figure 5: Per-dimension risk distribution on 27 real DID papers. The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large *unknown* share that marks retrieval failure rather than over-flagging.

Table 8: Deterministic calibration vs the human gold (55 cells), two policies for weak-retrieval *high*: demote to *medium* (keep coverage) or abstain to *unknown*. Both cut over-severity and preserve high recall. Pilot scale: only 2 gold cells are *high*.

metric	before	demote→med	abstain→unk
over-severe (answ.)	28	18	6
high prec. / rec.	0.04 / 0.50	0.50 / 0.50	0.50 / 0.50
exact agr. (answ.)	0.15	0.45	0.54
wt. κ (answ.)	0.06	0.13	0.25
answered / unk	33 / 22	33 / 22	13 / 42

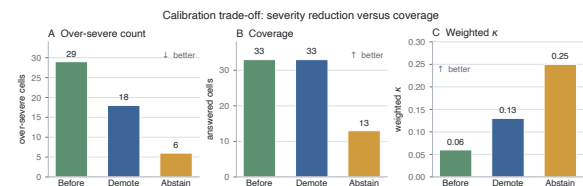


Figure 6: **Calibration trade-off: severity reduction versus coverage** against the adjudicated human gold. Demoting weak-retrieval *high*s to *medium* preserves coverage (33 answered) while cutting over-severe judgments from 29 to 18; abstaining gives the strongest correction (over-severe 6, weighted κ 0.25) but leaves only 13 answered cells. Calibration corrects severity bias; the remaining bottleneck is evidence grounding, not severity thresholding alone.